

Three Generations of Healthcare IT: From the Digital Record to the Computable Care Process

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Abstract

Healthcare information technology can be read as a progression in the *unit of information it makes computable*. The first generation made the clinical **record** computable, replacing paper charts with electronic documents and transactional databases. The second generation made the clinical **fact** computable, using controlled terminologies, structured systems, interoperability standards, and clinical natural-language processing to encode diagnoses, medications, procedures, and results as computable clinical state. We argue that an emerging **third generation makes patient-specific clinical intent computable from clinical communication**: what should happen next, when, under which conditions, and by whom, information that today lives in free-form notes, discharge instructions, referrals, handovers, and messages. The claim is deliberately narrow. Existing standards and workflow systems, FHIR *CarePlan*, *ServiceRequest*, *Task*, and *PlanDefinition*, computer-interpretable guidelines, and process-mining pipelines, can already represent orders, tasks, care plans, and executable protocols *once these have been explicitly encoded*. The unresolved gap is upstream: much patient-specific intent is expressed only in natural communication and therefore never enters this computable process layer. We define the **Actionable Clinical Record (ACR)** as the atomic object of that layer: a source-grounded representation of an intended clinical action together with its temporal, conditional, responsibility, dependency, status, and confidence semantics. Modern AI, large language models (LLMs) for semantic interpretation coupled with symbolic reasoning for verifiable structure, is what finally makes ACR recovery tractable, and it positions existing workflow infrastructure as the downstream machinery the third generation can at last populate automatically. We give explicit criteria for a generation, distinguish prescribed, observed, and intended process, define an ACR maturity ladder and an executable-correctness benchmark suite, and state reliability as a formal system property rather than a slogan. We ground the argument in a working instance, reliable follow-up extraction, where a hybrid neural-symbolic pipeline reaches action-time Pair F1 of 0.997 on a controlled benchmark while general-purpose LLMs reach about 0.53, and set out testable predictions and the Computable Care Process research agenda.

Keywords: healthcare IT; electronic health records; unit of computability; actionable clinical record; clinical intent; large language models; neural-symbolic AI; temporal reasoning; care-process automation; FHIR workflow

1. Introduction

Healthcare information technology is best read not through the technologies that happen to be fashionable in a given decade but through the *unit of information it makes computable*. Technologies turn over quickly; the computational object persists. Organized this way, the field's history is a single sentence: **the first generation made clinical information available to computers; the second made clinical state interpretable by computers; the third makes clinical intent operational by computers**. The first made the *record* computable; the second made the *fact* computable; the third, we argue, makes patient-specific *clinical intent* computable from communication (Figure 1). Generational language has been used before for electronic medical records and for the twenty-five-year

evolution of health IT,^[1,2] but the organizing principle we propose, the unit of computability paired with its enabling computational stack, is a synthesis rather than a restatement: each transition unlocked a distinct class of computation, required a distinct stack, and served a distinct set of clinical needs, and each left a residue that the next generation had to address.

The residue left by the first two generations is the same one clinicians experience daily. A physician writes “repeat the CBC in two weeks” or “refer to cardiology if symptoms persist,” and that instruction, though clinically decisive, is captured only as narrative text. The record stores it (generation one) and may even code the associated diagnosis or medication (generation two), but the *action*, its *timing*, its *condition*, and its *owner* remain outside the computable substrate. The consequence is measurable: missed test-

result follow-up ranges from 6.8% to 62%,^[3] roughly half of recommended imaging follow-ups are never performed,^[4] and only about a third of referrals reach a completed specialist visit.^[5]

What the claim is, and is not. It would be wrong to say healthcare IT could not represent process before LLMs. It could, whenever a person or system had already structured that process. FHIR represents patient-specific intent through *CarePlan* and *ServiceRequest* and generic plans through *PlanDefinition*;^[15] computer-interpretable guidelines have encoded executable protocols for decades;^[35,36] and process-mining pipelines reconstruct patient pathways from event logs.^[42] These are not competitors to the third generation but its downstream infrastructure. It helps to separate three fundamentally different kinds of process information: *prescribed* process (“patients with X should undergo Y”), the domain of guidelines; *observed* process (“the patient actually underwent $A \rightarrow B \rightarrow C$ ”), the domain of records and process mining; and *intended* process (“for *this* patient, do X in two weeks, unless Y”), which is communicated during care. Guidelines address the first, process mining the second. The third generation targets the third, and the de-

fining problem is not generating clinical text but **transforming what people communicate about future care into source-grounded, verifiable, executable, and monitorable clinical commitments.**

Contributions. (1) explicit criteria for what constitutes a generation of healthcare IT, and a three-generation model organized by the unit of computability and its enabling computational stack; (2) a formal definition of the *Actionable Clinical Record* (ACR), the atomic object the third generation produces, with a maturity ladder from *mentioned* to *closed*; (3) the transformation Communication $\rightarrow$ ACR $\rightarrow$ workflow, positioning the ACR as the missing middle between natural clinical communication and existing structured infrastructure; (4) reliability stated as a formal system property, and an executable-correctness benchmark suite; (5) a working instance with empirical evidence, convergent evidence from adjacent systems, testable predictions, and the Computable Care Process research agenda.

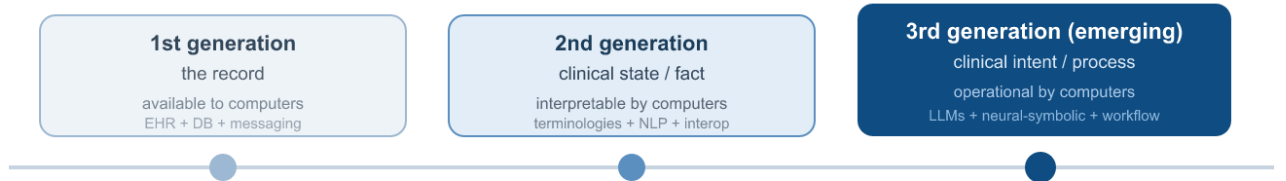


Figure 1. Three generations of healthcare IT, organized by the unit of information each makes computable and the enabling computational stack. The bottom row is the paper’s shorthand: each generation answers a different question.

2. What Defines a Generation of Healthcare IT

A generation is not any trend; it is a shift with structural consequences. We call a shift a generation when it satisfies five markers: (i) a new **atomic unit of computability**; (ii) a **driving force** that makes the shift economically or clinically inevitable; (iii) an **enabling technology and standard**; (iv) a new **class of computation** the unit unlocks; and (v) a **residual limitation** that the shift cannot solve and that seeds the next generation. This makes the framing checkable: a purported generation that fails these markers is a fea-

ture, not a generation, and the model predicts that the residual limitation of one generation names the unit of the next. Table 1 shows all three generations satisfying the markers, with the residue of each pointing at its successor. The markers are facets of a single shift rather than five independent tests; their force is the descriptive coherence they produce together with one predictive claim, that the residual limitation of each generation names the unit of the next.

Marker	Gen 1: Record	Gen 2: Facts	Gen 3: Care process
Unit of computability	Clinical document	Discrete clinical fact	Intended clinical action (the ACR)
Driving force	HITECH / Meaningful Use policy and reimbursement	Interoperability and quality-measurement mandates	Maturation of clinical NLP and LLMs; care-continuity and safety pressure
Class of computation unlocked	Store, retrieve, exchange documents	Compute <i>on</i> facts (decision support, quality, research)	Compute <i>on</i> process (monitor, prioritize, close loops)
Residual limitation (seeds next)	Content is unstructured narrative	Facts are coded, but actions and process are not	Reliability, safety, evaluation, integration (open)

Related framings. Our lens is complementary to existing models, not a competitor. The data-information-knowledge hierarchy (DIKW) describes a generic ascent from raw data to actionable knowledge;^[6] we specialize the question to *which clinical object* is made computable at each step. Maturity models such as the HIMSS Electronic Medical Record Adoption Model (EMRAM), an eight-stage ladder, and the wider family of hospital digital-maturity frameworks measure *how much* capability an institution has adopted;^[7,8] our model is orthogonal, naming *what kind* of information becomes computable, with the third generation lying beyond the capabilities those ladders currently score. Prior work has also used generational and evolutionary language for electronic health records;^[1,2] we make the organizing principle explicit and testable.

Scope of the model. The three generations describe a progression of *capabilities*, not a fixed global timeline: adoption differs by country and institution, and the periodization is anchored in the well-documented history of United States electronic health records. The generations are cumulative, each layer subsuming the ones beneath it. The empirical grounding offered here is a single feasibility instance (Section 8); the model's value is as an organizing lens and a source of testable predictions.

3. Generation One: Digitizing the Record

The first generation replaced the paper chart with an electronic one. Its defining achievement was capture, storage, retrieval, and transmission of clinical documents, driven in the United States by the HITECH Act (2009) and the Meaningful Use incentive program, which moved EHR adoption from a minority of hospitals to near-universal within a decade.^[9,10] The intellectual root is older: Weed's problem-oriented medical record proposed that the chart be an organized, auditable information structure rather than a chronological narrative.^[11] Its enabling technology was the transactional database, with no automated interpretation of clinical content.

Generation 1: concrete examples

Systems: Epic, Oracle Health/Cerner, MEDITECH, Allscripts; the U.S. Department of Veterans Affairs VistA/CPRS; PACS for radiology images; departmental LIS and pharmacy systems.

Algorithms & tech stack: relational databases and transactional (OLTP) systems; document imaging and scanning; structured forms and templates; client-server and later web architectures; HL7 v2 messaging for point-to-point exchange; DICOM for images.

Use cases & needs: legibility and availability of the chart; e-prescribing and order entry (CPOE); charge capture and billing; regulatory reporting; basic clinical documentation. The unmet need it exposed: the

record was digital but its *content* was largely unstructured narrative.

4. Generation Two: Digitizing Clinical Facts (Computable Clinical State)

The second generation made discrete clinical facts computable and exchangeable, turning the record into computable clinical *state*. Its enabling stack was broad, and it is a mistake to reduce it to one technique: structured data entry and templates, controlled terminologies and ontologies, computerized order entry and coded problem and medication lists, clinical data warehouses, and interoperability standards were at least as important as language processing.^[14,15] Clinical natural-language processing was *one route* by which narrative facts entered this semantic layer: systems such as MedLEE and cTAKES parsed reports into coded concepts,^[12,13] complementing facts that were captured as structured data at the point of entry. Together these enabled the first broad wave of computation *on* clinical data: decision support, quality measurement, registries, and secondary research use.^[16] Yet a persistent tension appeared: forcing clinical meaning into structured fields captures some facts while much of the reasoning, and nearly all of the intended *action*, stays in narrative.^[17]

Generation 2: concrete examples

Systems: structured EHR modules and clinical decision support (CDS); clinical-NLP pipelines (MedLEE, cTAKES, CLAMP); health information exchanges (HIEs); SMART on FHIR apps and terminology servers;^[18] clinical data warehouses and research networks such as i2b2 and OHDSI/OMOP; eCQM quality-measure engines.

Algorithms & tech stack: rule-based and statistical clinical NLP for concept extraction; controlled terminologies and ontologies (SNOMED CT, ICD-10, LOINC, RxNorm);^[14] rule-based CDS (drug-interaction checks, alerts, Arden syntax); HL7 v2/v3, CDA, and FHIR for interoperability;^[15] the OMOP common data model and ETL; structured data entry; phenotyping and supervised ML on tabular EHR data.

Use cases & needs: interoperability and data exchange; decision support and safety alerting; quality and performance measurement; population health and registries; secondary use for research;^[16] coded problem/medication lists and reimbursement coding. The unmet need it exposed: coding a *fact* is not the same as capturing an *action*; the care process, distributed across communication, remained outside the structured layer.

5. The Gap: The Care Process Lives in Communication

Both prior generations optimized *state*, the record and the facts it contains, but not *process*, the sequence of intended actions that constitutes care. Follow-up in-

structions, conditional plans, handover items, and self-care directives are authored as language and consumed as language, and they fail as language: they are forgotten, mis-scheduled, or never closed.^[3,17] The clinical-safety literature documents this loop-closure failure as a driver of diagnostic error and avoidable harm, with system-level economic weight.^[19,20] The gap is structural, not incidental: the missing information is an *executable representation of what should happen next*, and neither the digital document nor the coded fact encodes it. A single note makes the three generations concrete:

One note, three generations

Clinical note: “Repeat CBC in two weeks; refer to cardiology if chest pain persists.”

Generation 1 (record): the sentence is stored verbatim as searchable text; nothing about it is computable beyond keyword search.

Generation 2 (facts): “CBC” maps to a LOINC laboratory concept and “chest pain” to a SNOMED problem, but the *plan*, the repeat, the referral, and their timing and condition, stays in narrative.

Generation 3 (clinical intent): two Actionable Clinical Records are emitted, each grounded in its source span and monitorable:

{action: repeat CBC, temporal: +14 days, status: pending, provenance: span#1}

{action: cardiology referral, condition: chest pain persists, actor: physician, status: conditional, provenance: span#2}

6. Generation Three: Making Clinical Intent Computable from Communication

The third generation makes patient-specific clinical intent computable by recovering it from clinical communication and emitting it as structured, executable records. Its enabling stack is modern AI: large language models for semantic interpretation, coupled with symbolic reasoning for verifiable structure. The atomic object is no longer a document or a fact but the **Actionable Clinical Record (ACR)**: a source-grounded, normalized statement of an intended clinical action together with the semantics needed to schedule, assign, monitor, and audit it. We define it formally below, because a named, shared object is what lets others

build on the framework, extending the schema, evaluating ACR recovery, or mapping ACRs to workflow, rather than agreeing with an essay.

6.1 The Actionable Clinical Record, defined

An ACR is a tuple

ACR = (action, target, actor, temporal constraint, condition, dependency, status, provenance, confidence)

with **required** fields *action*, *status*, and *provenance*, the minimum for a record that is safe to act on and to audit, and **optional** fields that are present when the communication supports them. The fields are defined semantically, not as free text:

ACR field semantics

action the intended clinical act (e.g. repeat a test, refer, adjust a medication, escalate). **target** the object of the action (the CBC, cardiology, the drug and new dose), linked where possible to a Gen-2 coded concept.

actor the responsible party (ordering physician, specialty service, patient, nurse); may be unresolved and flagged as such. **temporal constraint** a normalized time semantics, not merely a date: absolute (“within six months”), relative (“before the next visit”), event-anchored (“after finishing antibiotics”), or conditional-recurring, any of which may resolve to a computed due date.

condition the clinical trigger or precondition (“if chest pain persists”). **dependency** ordering or prerequisite relations to other ACRs or results. **status** position on the maturity ladder below (mentioned...closed).

provenance (required) the grounding of the record: source communication, source span, speaker, and encounter, so every executable item traces to the exact text that produced it. **confidence** a calibrated, per-field uncertainty used to route uncertain records to human review.

Concretely, the representation extends from the simple *Action* + *Time* pair proven in Section 8 to this full tuple (Figure 2), with provenance and confidence present on every record:

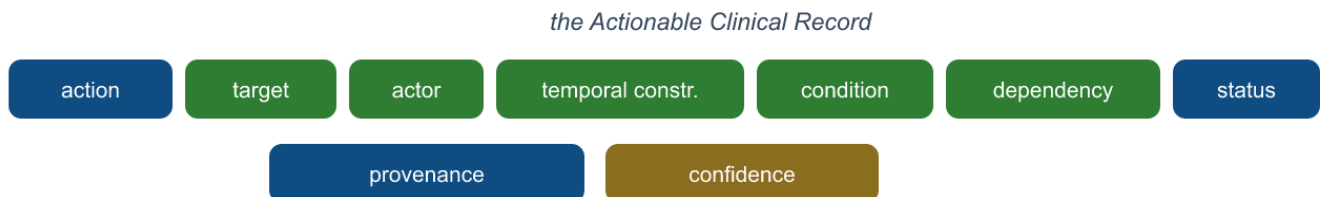


Figure 2. The ACR schema. *action*, *status*, and *provenance* are required; the remaining fields are populated when the communication supports them. Every record carries a per-field confidence used for selective human review.

6.2 The ACR maturity ladder

“Generation 3” is not a single capability but a ladder, and separating its rungs prevents both over- and under-claiming. Recognizing that “CBC” appears in a note is Gen 2. Extracting *repeat CBC* as an intended action is only partial Gen 3. Linking that action to “two weeks” and to a responsible actor produces an ACR; mapping it to a computed due date and a FHIR object makes it executable; detecting completion or escalation

closes the loop. We name the rungs **mentioned** → **interpreted** → **actionable** → **executable** → **monitored** → **closed**, which also draws the distinction between an *actionable* record (a well-formed intent) and an *executable* one (an intent with enough resolved information to act on automatically). Systems can be classified by the highest rung they reach reliably, and the same ladder doubles as the *status* field of the ACR.



Figure 3. The Gen-3 maturity ladder. Recognition of concepts is Gen 2; the third generation begins at interpreted intent and matures toward closed-loop execution.

Generation 3: concrete examples

Systems: ambient clinical documentation / AI scribes (Nuance DAX Copilot, Abridge, Suki, Nabla) that turn the spoken encounter into notes;^[21] follow-up and care-gap engines; referral-loop and results-management tools; agentic clinical-workflow assistants; the MedFollow / CareLoop line of work as a research instance.^[22]

Algorithms & tech stack: clinical language models (BioBERT, ClinicalBERT, GatorTron, MedPaLM);^[23–26] clinical information extraction (MedLEE, cTAKES, CLAMP) inherited from the previous generation;^[12,13,27] temporal extraction and normalization (i2b2-2012 temporal relations, THYME / Clinical TempEval, HeidelTime / SUTime);^[28–32] neural-symbolic pipelines pairing learned extraction with deterministic temporal reasoning;^[33] selective prediction and uncertainty calibration; event and relation extraction;^[34] FHIR write-back for actionable outputs.^[15]

Use cases & needs: follow-up instruction extraction; care-gap and referral-loop closure; scheduling and reminder automation; quality and safety monitoring; safe handovers; patient-message triage; turning

ambient-captured encounters into executable tasks. Needs it imposes: reliability over fluency, correct temporal computation, source-linked auditability, and human oversight for uncertain cases.

6.3 From communication to workflow: the missing middle

The third generation is best understood as a bridge, not a replacement EHR. Its input surface is the full range of clinical communication, discharge instructions, patient–clinician conversation, EMS and nurse handovers, referral correspondence, portal messages, telephone calls, and ambient encounter transcripts. Its output is a layer of ACRs that maps onto infrastructure the field already has: FHIR *Task*, *ServiceRequest*, and *CarePlan*, schedulers, alerts and work queues, referral systems, and results management.^[15] Healthcare already knows how to *execute* known instructions; the unmet need is making instructions *known to the machine* when they originate as natural communication (Figure 4).

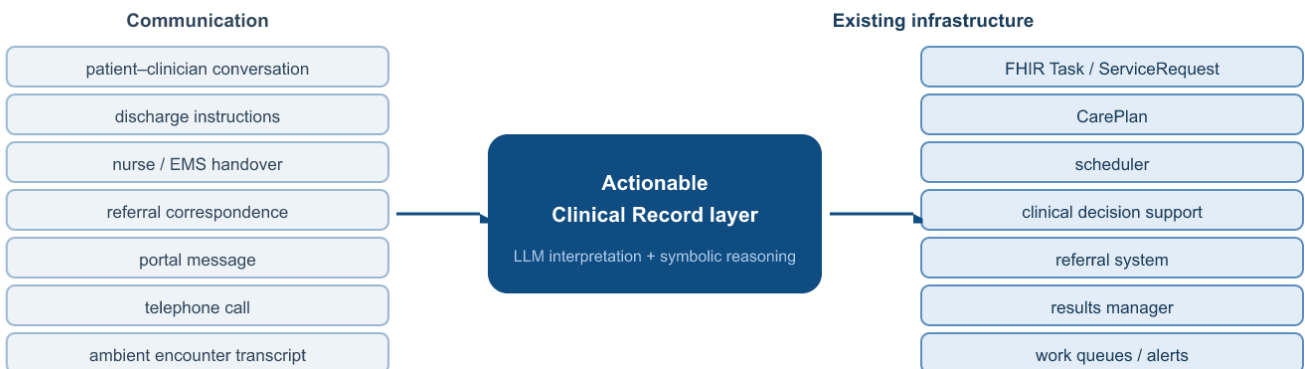


Figure 4. The missing middle. Healthcare already has infrastructure to execute known instructions (right); the ACR layer (center) recovers patient-specific intent from natural communication (left) so it can reach that infrastructure automatically.

Follow-up is the most obvious case because its safety and economic consequences are documented, but it is one member of a category. Table 2 lists communica-

tion-derived process objects that the same ACR machinery targets.

Process object	Example communication	Salient ACR fields
Follow-up	“Repeat CBC in two weeks.”	action, temporal constraint (+14d), actor
Conditional escalation	“Return immediately if fever develops.”	action, condition, actor=patient
Medication transition	“Reduce the dose after seven days.”	action, target (drug/dose), temporal constraint
Handoff commitment	“Cardiology to review the echo tomorrow.”	action, actor=service, temporal constraint
Referral	“Refer to dermatology for the lesion.”	action, target, actor, dependency
Pending-result responsibility	“I will follow up on the biopsy result.”	action, actor, dependency (on result)
Self-management action	“Check blood pressure daily and log it.”	action, actor=patient, recurring temporal
Patient-declared intent	“I will stop taking it until I speak with my doctor.”	action, actor=patient, condition

6.4 What the third generation is not

Two mature lines can each be mistaken for the third generation, and the distinction sharpens the claim. **It is not classical clinical NLP.** Concept extraction from notes has existed for decades (MedLEE, cTAKES, CLAMP), but it produces *facts*: it records that “MRI” appears, feeding the second-generation structured layer.^[12,13,27] The third generation produces an *executable action*: that an MRI is due in two weeks, still pending, owned by a specific actor, at a reliability high enough to act on rather than only index. The object is different (process, not fact), the bar is different (act, not retrieve), and LLM-era semantics are what finally make it tractable at scale. Clinical NLP improving is therefore not, on its own, a generational shift: generations here are cumulative layers, and the third is distinguished by the object it makes computable and the reliability bar it must clear, not by the raw technique. **It is not computer-interpretable guidelines.** That program encoded *generic, top-down, population-level* protocols as executable models.^[35,36] The third generation extracts *patient-specific, bottom-up* actionable records from the actual communication of a given encounter. The two are complementary: guidelines say what should generally happen; the third generation captures what a clinician actually instructed for this patient.

6.5 Convergent evidence that the generation is emerging

The claim does not rest on a single system. Independent developments are converging on the actionable record. **Ambient documentation** has moved into routine practice, converting the spoken encounter into notes and, increasingly, into orders and tasks.^[21,37] **Clinical LLM evaluation** is maturing toward task competence rather than fluency: benchmarks now measure diagnostic-questioning efficiency in patient

conversations, treating the clinical dialogue itself as a computable, measurable process.^[38] **Care-gap, results-management, and referral-loop tools** already act on structured follow-up signals, and early **agentic clinical assistants** attempt multi-step workflow actions. These are separate research and product lines, blind to each other, arriving at the same object: a structured, monitorable representation of what should happen next. MedFollow is our controlled feasibility proof within this convergence (Section 8).

6.6 Will end-to-end LLMs make the symbolic layer obsolete?

A more capable generator does not remove the requirements the third generation imposes. Auditability, verifiable temporal correctness, and a calibrated signal of when to defer are properties of the *output contract*, not gaps in model capability: a system that acts on patient trajectories must prove that a date is correct and trace it to its source, however fluent the underlying model becomes. The symbolic layer supplies guarantees, not missing knowledge, which is why reliability, not scale, is the binding constraint.^[38,39]

7. A Comparative View of the Three Generations

Table 3 consolidates the three generations. The top rows make the organizing principle memorable, each generation answers a different question and emits a different canonical object, while the lower rows ground it in systems, stack, and value. The row we would ask readers to remember is *Question: What was documented?* → *What is true?* → *What should happen?*

Axis	Gen 1: Record	Gen 2: Facts	Gen 3: Care process
Object	Record	Clinical state / fact	Clinical intent / process
Question	What was document-	What is true about the pa-	What is supposed to happen?

Axis	Gen 1: Record	Gen 2: Facts	Gen 3: Care process
	ed?	tient?	
Representation	Documents / events	Concepts / observations	Actions, constraints, dependencies
Primary computation	Store / retrieve / exchange	Query / reason / predict	Schedule / coordinate / monitor / close
Enabling computational stack	EHR + relational/OLTP DB + HL7 v2 + DICOM + messaging	Terminologies + structured systems + interoperability + NLP + ML on tabular data	LLM/ML + relation & temporal reasoning + workflow semantics + selective prediction
Representative systems	Epic, Cerner, VistA/CPRS, MED-ITECH; PACS, LIS	CDS, clinical-NLP pipelines, HIE, SMART-on-FHIR apps, i2b2/OMOP warehouses, eCQM engines	Ambient scribes (DAX, Abridge, Suki, Nabla); follow-up/care-gap engines; agentic workflow assistants; MedFollow/CareLoop
Canonical output	Document	Fact	Actionable Clinical Record
Value unlocked	A legible, available digital substrate; billing and basic safety	Computation on data: decision support, quality measurement, population health and research	Loop closure and follow-up safety; care-coordination and workflow automation
Failure residue	Meaning trapped in narrative	Intent trapped in communication	Safe, reliable execution

8. A Working Instance: Reliable Follow-Up Extraction

Follow-up extraction is the smallest complete instance of the third generation: it takes clinical communication and emits an executable Action + Time record. Our MedFollow study is a feasibility proof that the instance is buildable and that architecture is decisive.^[22] A hy-

brid neural-symbolic pipeline, learned extraction of actions and time expressions coupled with deterministic temporal normalization, was evaluated on a controlled 2,000-note benchmark against generative baselines.

Metric	Value
Pair F1, seen actions (hybrid)	0.997
Pair F1, unseen action types (hybrid)	0.986
day temporal error (hybrid)	0.00
Pair F1, GPT-4o-mini / LLaMA-3	~0.53

The generative models recognized actions accurately (action F1 above 0.96) yet collapsed on the end-to-end task, because linking and temporal reasoning remained implicit in generation. The lesson generalizes to the whole third generation: the value is created not in recognizing clinical concepts but in *reliably assembling them into an executable representation*, which is where explicit, verifiable structure earns its place. We are explicit about the result’s standing: it is one controlled, synthetic-benchmark instance, and its role is feasibility, not deployment coverage. The framework does not rest on the number 0.997; that figure demonstrates only that the smallest complete ACR (Action + Time) is buildable and that architecture is decisive. The generational thesis rests instead on the object (the ACR) and the evaluation that measures its recovery.

8.1 An executable-correctness benchmark suite

Because the third generation acts, it must be evaluated on *executable correctness* rather than text overlap. We propose a benchmark suite for ACR recovery, each dimension a separate, reportable measure: **action detection** (are intended actions found?); **argument**

extraction (are target, actor, condition recovered?); **action-time / condition / actor linking** (are arguments bound to the right action?); **temporal-normalization error** (days between computed and reference due dates); **executable-record exact match** (whole-ACR correctness); **unsupported-action rate** (fabricated commitments, ideally zero); **omitted-action rate** (missed commitments); **calibration and selective-risk curves** (does confidence predict correctness, and does abstention improve accuracy on the accepted set?); **source-provenance accuracy** (does each record point to the correct span?); and, ultimately, **loop-closure outcome** (does the acted-upon record reach completion?). The extension from Action + Time to the full ACR tuple is the natural next increment, motivated by real transcribed notes containing medication changes, conditional and generic follow-up, self-care, and actions without explicit timing.

9. Enabling Technologies and Requirements for the Third Generation

9.1 Enabling technologies

The third generation stands on four technical pillars. **Clinical language models** (BioBERT, ClinicalBERT, GatorTron, and large medical LLMs such as Med-PaLM) supply robust semantic interpretation of heterogeneous clinical language.^[23–26] **Clinical information extraction**, the NLP line inherited from the previous generation (MedLEE, cTAKES, CLAMP, and the i2b2/n2c2 benchmarks), supplies the entities and relations.^[12,13,27,34] **Temporal reasoning**, from the i2b2 2012 temporal-relations challenge and Clinical TempEval to normalization engines, supplies the calendar arithmetic that turns “in two weeks” into a date.^[28–32] **Ambient documentation** extends the input surface from written notes to the spoken encounter.^[21,37]

9.2 Why fluent generation is not enough

General-purpose large language models interpret clinical language well but are unreliable where the third generation is most demanding: linking an action to its time and normalizing it into an executable schedule, without omission or fabrication, and with a calibrated signal of when to defer.^[39,40] Reliability, not fluency, is the binding constraint. This motivates **neural-symbolic** designs that pair learned extraction with deterministic, verifiable reasoning,^[33] and **selective prediction** that routes uncertain cases to a human.

9.3 Reliability as a formal system property

“Reliability, not fluency, is the binding constraint” should be read as an architecture specification, not a slogan. We state it as a set of invariants a care-process system must satisfy, each checkable and each shaping the design rather than decorating it: **(I1) no executable item without provenance**, every acted-upon ACR traces to a source span, speaker, and encounter; **(I2) deterministic computation where deterministic semantics exist**, calendar arithmetic and dependency ordering are computed, not generated; **(I3) uncertainty on every inferred field**, a calibrated confidence accompanies each value; **(I4) abstention / selective prediction**, the system may decline rather than guess; **(I5) risk-proportionate human review**, higher-consequence actions demand tighter oversight; **(I6) an immutable trace from communication to action**, so any executed commitment can be reconstructed and audited;^[41] and **(I7) an explicit boundary between extracted and inferred content**, what the clinician actually said versus what the system concluded. This is also where the neural-symbolic argument becomes precise: the point is not that symbolic AI is intrinsically superior, but that some workflow semantics (dates, dependencies, status transitions) are externally verifiable, and verifiable semantics should not remain latent inside generative in-

ference. The invariants connect the third generation to the older program of computer-interpretable guidelines and care pathways, which sought executable models of process but lacked a reliable path from free-text communication into those models.^[35,36]

10. Predictions and the Computable Care Process Research Agenda

10.1 Testable predictions

If the three-generation model is right, the following are checkable over the next several years, and their failure would count against it:

P1. The machine-actionable use of existing FHIR workflow resources (*Task*, *CarePlan*, *ServiceRequest*) will grow: today used mainly for documentation and exchange, they will increasingly carry executable, closed-loop follow-up, and the standard will extend their action and temporal semantics accordingly. *Checkable by* a rising share of follow-up actions represented as these resources rather than free text (horizon 3–5 years).

P2. Evaluation will shift from text-overlap metrics to *executable correctness*: action–time pair linking, temporal accuracy, and loop-closure rate. *Checkable by* new clinical benchmarks reporting these measures rather than ROUGE/BLEU-style overlap (horizon 2–4 years).

P3. Reliable systems will converge on *source-linked*, *auditable* action records with selective human review, rather than end-to-end free generation. *Checkable by* deployed follow-up and results-management tools exposing the source span and a confidence-based review queue (horizon 3–5 years).

P4. Ambient documentation will move downstream from producing notes to producing *tasks and orders*, closing the loop into scheduling and results management. *Checkable by* ambient-scribe products adding order and task generation and results follow-up, not just note drafting (horizon 2–4 years).

10.2 The Computable Care Process research agenda

Beyond the predictions, the framework defines a menu of publishable problems, a shared agenda a community can work through. We group them so that a system or study can locate itself precisely.

The object. A consensus *ACR ontology*, an extensible taxonomy of clinical actions, required-versus-optional field semantics, and reconciliation when messages conflict, revise, or revoke a plan. **Extraction.** ACR recovery from multilingual and multimodal communication (written notes, speech, ambient audio), transfer to unseen action types, and cross-message reconciliation that assembles one commitment from several exchanges. **Reasoning.** Temporal and causal reasoning over event-anchored and conditional constraints; distinguishing and reconciling *patient* intent from *clinician* intent. **Reliability.** Calibrated abstention, provenance verification, and the invariants of Sec-

tion 9 as measurable properties. **Evaluation.** Authentic-data benchmarks beyond synthetic corpora, and the executable-correctness suite of Section 8.1 adopted across tasks. **Integration and impact.** ACR → FHIR mappings and write-back into scheduling, results-management, and referral systems;^[15,35] human-factors design of the review queue; regulatory categorization of care-process software; and, the outcome that matters most, whether automatic loop closure measurably improves clinical outcomes. Moving from synthetic corpora to de-identified real notes across languages and documentation styles remains the central empirical risk, and safe deployment presupposes the selective prediction, auditability, and oversight named above.

11. Conclusion

The first generation made clinical information available to computers; the second made clinical state interpretable by computers; the third makes clinical intent operational by computers. Each is defined by the unit it makes computable, record, then fact, then intent, and by the computational stack that makes it so. The defining problem of the third generation is not generating clinical text but transforming what people communicate about future care into source-grounded, verifiable, executable, and monitorable clinical commitments, the Actionable Clinical Record. It does not replace the field's workflow infrastructure; it supplies the missing middle that finally lets natural clinical communication reach that infrastructure automatically, under reliability treated as a formal system property rather than a hope. Follow-up extraction shows the smallest complete instance is within reach today, convergent developments show it arriving from several directions at once, and the ACR schema, maturity ladder, benchmark suite, and predictions here are offered so that others can build on the framework rather than merely agree with it.

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